Bachelor's thesis

ENHANCING FAKE NEWS CLASSIFICATION WITH ADVANCED NLP MODELS

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Declaration

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Abstract

This study extends prior research in which relatively simple, classical machine learning models were employed to classify news articles or summaries as either fake or reliable. The objective of the present work is to investigate the use of more advanced architectures—specifically transformer-based models—to enhance classification performance on the same datasets utilized in the previous study.

Keywords transformers, BERT, DistillBERT, RoBERTa, GPT, fake news, deep learning, natural language processing, binary classification

Abstrakt

Tato studie navazuje na předchozí výzkum, ve kterém byly pro klasifikaci zpravodajských článků nebo jejich shrnutí jako falešných nebo důvěryhodných použity relativně jednoduché klasické modely strojového učení. Cílem této práce je prozkoumat využití pokročilejších architektur — konkrétně modelů založených na transformerech — za účelem zlepšení klasifikační výkonnosti na stejných datových sadách, které byly použity v předchozí studii.

Klíčová slova transformery, BERT, DistillBERT, RoBERTa, GPT falešné zprávy, hluboké učení, zpracování přirozeného jazyka, binární klasifikace

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List of abbreviations

BERT	Bidirectional Encoder Representations from Transformers
DistilBERT	Distilled BERT
RoBERTa	A Robustly Optimized BERT Pretraining Approach
GPT	Generative Pre-trained Transformer
NLP	Natural Language Processing

Chapter 1
Introduction

This thesis is primarily focused on the practical application of modern natural language processing methods for fake news classification. It builds upon the bachelor thesis by Jan Flajžík, Classification of Fake News in The Media/Social Media Ecosystem [1], which explored the use of classical machine learning models for predicting whether a news article is fake or reliable. The present work extends this research by employing more advanced transformer-based architectures.

The theoretical part of the thesis is limited mainly to the description of the selected models and their underlying principles. Rather than providing a deep theoretical analysis of the architectures, this work approaches the problem from a practical perspective by applying advanced models, evaluating their performance, and comparing the obtained results with those reported in previous work.

The datasets used in this work remain identical to those utilized in the previous work [1]. The primary difference lies in the use of computationally more demanding transformer models and in the integration of modern tools for experiment tracking and analysis, specifically the MLFlow framework.

The motivation for this research is well described in the previous work [1] and remains highly relevant today. The rapid spread of manipulative and misleading information through online platforms represents a significant societal issue. Due to the enormous volume of digital content published daily, manual verification of information is economically and practically infeasible. Consequently, automated fake news detection using machine learning techniques has become an important research area.

The following chapters introduce the selected transformer models, describe the implementation and experimental methodology, present the achieved results, and compare them with those obtained in previous research. Finally, the thesis discusses the limitations of the proposed approach and outlines possible directions for future work.

1.1 Goals

The primary objective of this thesis is to extend previous research in fake news classification by integrating advanced natural language processing architectures, particularly transformer-based models such as BERT, GPT, and related approaches. The work focuses on evaluating whether these modern architectures can improve classification performance compared to the classical machine learning methods used in earlier studies.

The thesis also examines different text preprocessing strategies and their influence on model performance. Experimental evaluation is conducted on the same datasets as those used in the previous work [1], enabling a direct comparison between traditional and transformer-based approaches.

The main goals of this thesis are as follows:

1. Investigate state-of-the-art methods for fake news classification.
2. Implement and evaluate at least three advanced transformer-based models, such as BERT, GPT, or related architectures.
3. Conduct experiments using different preprocessing techniques and compare the obtained results with those reported in previous work.
4. Analyze the strengths and limitations of the selected models and discuss possible improvements for future research.

Chapter 2

Used Methods

Before discussing the methods used in this work, it is useful to briefly review the models employed in the previous study by Jan Flajžík [1]:

- Naive Bayes
- Random Forest
- Convolutional Neural Network (specifically the OPCNN-FAKE architecture)
- Long Short-Term Memory (LSTM) network implemented as a simple one-layer neural architecture

One of the observations raised during the evaluation of the previous work was the absence of more advanced machine learning architectures. This thesis addresses that limitation by investigating the extent to which transformer-based models can improve fake news classification performance compared to the previously used classical approaches.

It is also important to note that the previous study achieved strong results, with accuracy exceeding 90% on English datasets and 80% on Czech-language datasets. In the results chapter, the performance of the transformer-based models will be compared with these previously reported results in order to evaluate whether the advanced architectures provide measurable improvements over the classical approaches.

2.1 Related Studies

Although this thesis primarily represents a direct continuation of the work by Jan Flajžík [1], additional recent studies related to fake news detection were also examined in order to better understand current trends and methodologies in the field.

The studies by [2] and [3] review recent developments in fake news classification and compare the performance of different machine learning and deep learning models across multiple datasets.

In particular, the Deep Learning for Fake News Detection by Haqi Al-Tai et al.[2] demonstrates that incorporating attention mechanisms can significantly improve the performance of classical neural architectures. The reviewed studies reported an accuracy of 45.3% on the LIAR dataset for CNN-based models enhanced with attention mechanisms, while transformer-based architectures achieved an accuracy of 40.55%. In the original LIAR dataset study by Wang, A New Benchmark Dataset for Fake News Detection [4], more traditional approaches achieved considerably lower results, with a classical CNN reaching 27% accuracy and LSTM architectures achieving 23.3%.

Although the LIAR dataset is not used in this thesis, these results suggest that attention-based architectures and transformer models generally outperform more traditional machine learning approaches. This observation further motivates the use of transformer-based models in the experiments conducted in this work.

2.2 Models

The experiments conducted in this thesis utilize the following transformer-based architectures:

- BERT
- DistilBERT
- RoBERTa
- GPT-2

The selection of these models was motivated by the intention to evaluate widely used and well-established transformer architectures while maintaining reasonable computational requirements. The chosen models represent relatively standard and lightweight transformer-based approaches that are suitable for experimentation under limited computational resources.

2.2.1 Transformers

All models used in this work are based on the Transformer architecture originally introduced in the paper Attention Is All You Need by Vaswani et al.[5]. Each selected model represents a modification or optimization of the original Transformer architecture and therefore exhibits different properties and performance characteristics.

The Transformer is a sequence transduction architecture that relies entirely on attention mechanisms, eliminating the need for recurrent or convolutional layers in order to model dependencies between input and output sequences. The original architecture follows an encoder-decoder design, where both components consist of stacks of identical layers containing multi-head self-attention mechanisms and position-wise feed-forward neural networks. Since the architecture does not utilize recurrent structures, positional encodings are incorporated into the input embeddings to preserve information about token order within the sequence. Residual connections and layer normalization are also applied throughout the network to stabilize and improve the training process.

Figure 2.1 illustrates the original Transformer architecture proposed in [5].

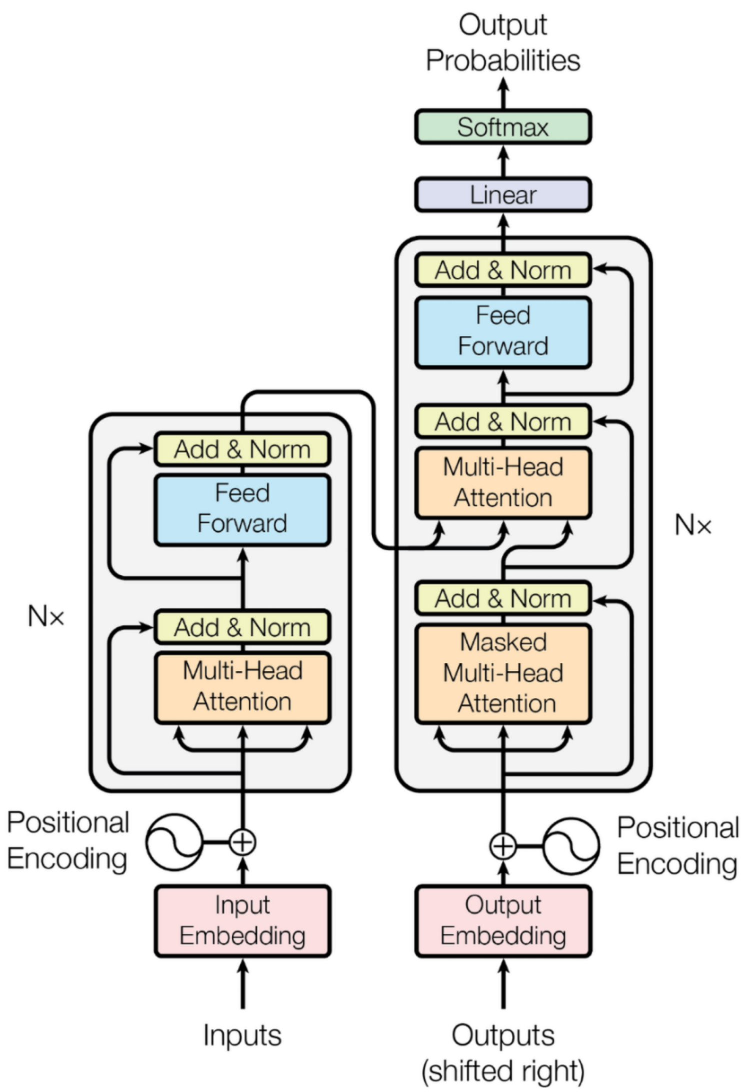


Figure 2.1 Original Transformer architecture introduced in [5]

The following subsections briefly describe the transformer-based models used in the experiments.

2.2.1.1 BERT

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding [6], is a language representation model designed to pre-train deep bidirectional contextual representations from large amounts of unlabeled text. Unlike traditional language models that process text in a single direction, BERT jointly considers both left and right context during training. This allows the model to capture richer contextual information and achieve state-of-the-art performance across a wide range of natural language processing tasks after fine-tuning.

BERT differs from the original Transformer architecture in several important aspects:

- Encoder-Only Architecture: While the original Transformer architecture follows an encoder-decoder structure intended primarily for sequence transduction tasks such as machine translation, BERT utilizes only the encoder component of the Transformer.
- Bidirectional Context Processing: Traditional autoregressive language models process text in a unidirectional manner, meaning that each token attends only to preceding tokens. In contrast, BERT employs bidirectional self-attention, allowing the model to simultaneously incorporate information from both preceding and following tokens.
- Masked Language Modeling (MLM): To enable bidirectional training, BERT introduces the Masked Language Modeling objective. During pretraining, approximately 15% of input tokens are randomly masked, and the model is trained to predict the original masked tokens using the surrounding context.
- Next Sentence Prediction (NSP): BERT additionally introduces the Next Sentence Prediction task, where the model learns relationships between pairs of sentences by predicting whether one sentence logically follows another in the original text. This objective improves performance on tasks involving sentence relationships, such as question answering and natural language inference.
- Input Representation: BERT uses a unified token representation capable of processing both single sentences and sentence pairs. Special tokens such as `[CLS]` and `[SEP]` are used to represent the entire sequence and to separate multiple sentences within the input.

2.2.1.2 DistilBERT

DistilBERT, introduced by Sanh et al., DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter [7], is a smaller and computationally more efficient variant of the BERT architecture. The model was designed to preserve most of BERT's language understanding capabilities while significantly reducing model size and training and inference time. DistilBERT is trained using the technique of knowledge distillation, where a smaller student model learns to reproduce the behavior of a larger teacher model, in this case BERT.

DistilBERT differs from the original Transformer architecture and from BERT in several important ways:

- Encoder-Only Architecture: Similar to BERT, DistilBERT uses only the encoder component of the original Transformer architecture.
- Reduced Model Complexity: DistilBERT contains approximately half the number of layers used in BERT, making the model substantially smaller and faster. According to the original paper, the model is approximately 40% smaller and 60% faster while retaining around 97% of BERT's performance.
- Knowledge Distillation Training: Instead of training the model entirely from scratch, DistilBERT is trained using a distillation objective that transfers knowledge from the larger BERT model. The training objective combines language modeling loss, distillation loss, and cosine embedding loss.
- Simplified Architecture: To further reduce computational complexity, DistilBERT removes several components present in BERT, including token-type embeddings and the pooler layer.
- Initialization Strategy: The model initialization process reuses selected layers from the original BERT model, allowing DistilBERT to benefit from the pretrained representations of the larger architecture.

2.2.1.3 RoBERTa

RoBERTa (Robustly Optimized BERT Pretraining Approach), introduced by Liu et al. [8], is an optimized variant of the BERT architecture. The authors of RoBERTa demonstrated that the original BERT model was significantly undertrained and that its performance could be improved through modifications to the training procedure and the use of substantially larger training datasets.

Like BERT, RoBERTa is based on the Transformer encoder architecture; however, it introduces several important modifications to the original training methodology:

- Encoder-Only Architecture: Similar to BERT, RoBERTa utilizes only the encoder component of the original Transformer architecture.
- Dynamic Masking: Unlike BERT's static masking strategy, RoBERTa applies dynamic masking, meaning that masked tokens are selected differently

each time a training sequence is processed. This increases variability during training and improves generalization.

- Removal of Next Sentence Prediction (NSP): RoBERTa removes the Next Sentence Prediction objective used in BERT. The authors found that eliminating this task and training on continuous text sequences led to improved model performance.

- Larger-Scale Training: RoBERTa was trained on significantly larger datasets and with substantially larger batch sizes compared to BERT. The training corpus exceeded 160 GB of text data, allowing the model to learn richer language representations.

- Byte-Pair Encoding (BPE): The model employs byte-level Byte-Pair Encoding tokenization with a large subword vocabulary, enabling efficient representation of text without relying on unknown tokens.

- Training on Longer Sequences: RoBERTa is trained primarily on full-length sequences of up to 512 tokens, which improves the model's ability to capture long-range contextual dependencies.

2.2.1.4 GPT-2

GPT-2 (Generative Pre-trained Transformer 2), introduced by Radford et al., Language Models are Unsupervised Multitask Learners [9], is a large-scale language model based on the Transformer decoder architecture. Unlike encoder-based models such as BERT and RoBERTa, GPT-2 follows a decoder-only autoregressive design, where the model predicts the next token based on the preceding context. The architecture was designed as a general-purpose language model capable of performing multiple natural language processing tasks without task-specific modifications.

The original GPT-2 model contains up to 1.5 billion parameters and demonstrated that sufficiently large language models can perform a variety of tasks, including text generation, summarization, translation, and question answering, in a zero-shot setting.

Several important characteristics distinguish GPT-2 from previous Transformer-based architectures:

- Decoder-Only Architecture: GPT-2 uses only the decoder component of the original Transformer architecture and processes text autoregressively from left to right.

- Large-Scale Training Dataset: The model was trained on the WebText dataset, which consists of millions of documents collected from outbound Reddit links with high user engagement. The resulting dataset contains approximately 40 GB of text data.

- Architecture Improvements: Compared to the original GPT model, GPT-2 introduces modifications such as adjusted layer normalization placement, an expanded context window of up to 1024 tokens, and a larger vocabulary size.
- Byte-Pair Encoding (BPE): GPT-2 employs byte-level Byte-Pair Encoding tokenization, allowing efficient representation of arbitrary Unicode text while avoiding problems related to unknown tokens.
- Zero-Shot Learning Capability: One of the most significant contributions of GPT-2 is its ability to perform multiple tasks without explicit task-specific fine-tuning. The model achieved strong performance on several benchmark datasets using only natural language prompts.
- Scaling Effects: The GPT-2 study demonstrated that increasing model size and training data significantly improves performance across a wide range of language modeling tasks. The authors also observed that even the largest GPT-2 variant still underfit the training dataset, suggesting that additional scaling could further improve performance.

Although GPT-2 demonstrated impressive text generation capabilities, the authors noted that some complex tasks still required further improvements before the model could be reliably applied in practical real-world scenarios.

2.3 Text Preprocessing Techniques

Similarly to the previous work by Jan Flajžík [1], several text preprocessing techniques were applied in this thesis. These preprocessing methods were treated as configurable components of the training pipeline and were evaluated in combination with different models and datasets.

The preprocessing techniques used in this work include:

- stemming,
- lemmatization,
- text cleaning operations such as punctuation removal, number removal, URL removal, and special character removal,
- tokenization using NLTK, BERT, and GPT-2 tokenizers,
- sequence transformation techniques such as token truncation.

Most preprocessing techniques were adapted from the implementation provided in [1]. However, the preprocessing pipeline was redesigned and extended to support transformer-based architectures and their corresponding tokenization methods. The BERT model-based and GPT-2 tokenizers were implemented using the libraries provided within the Hugging Face Transformers framework.

Chapter 3
Datasets

The experiments conducted in this thesis use the same datasets as those utilized in the previous work by Jan Flajžík [1]. Three of the datasets contain English-language texts, while one dataset consists of Czech-language texts.

Some preprocessing techniques from the original implementation, such as noise reduction, stemming, and lemmatization, were reused and adapted for the transformer-based pipeline used in this work. At the same time, additional tokenization approaches suitable for transformer architectures were introduced.

For all datasets used in this thesis, the target labels follow the same convention: - label `0` represents fake (unreliable news), - label `1` represents reliable news.

3.1 ReCOVery Dataset

The ReCOVery dataset was created following the COVID-19 pandemic and is intended for research focused on the credibility of COVID-19-related news articles. The dataset contains English-language articles discussing both the pandemic itself and its political and social consequences.

The original ReCOVery dataset is multimodal. However, in the previous work on which this thesis builds, only textual information was used. The same approach is adopted in this thesis. All columns except the article title and textual content were removed, and the remaining textual information was processed as a single input sequence.

The ReCOVery dataset is notably imbalanced, as illustrated in Figure 3.1. Although this imbalance was acknowledged in the previous work, no balancing techniques were applied. The same approach is followed in this thesis, and the imbalance should therefore be considered when interpreting the experimental results [1].

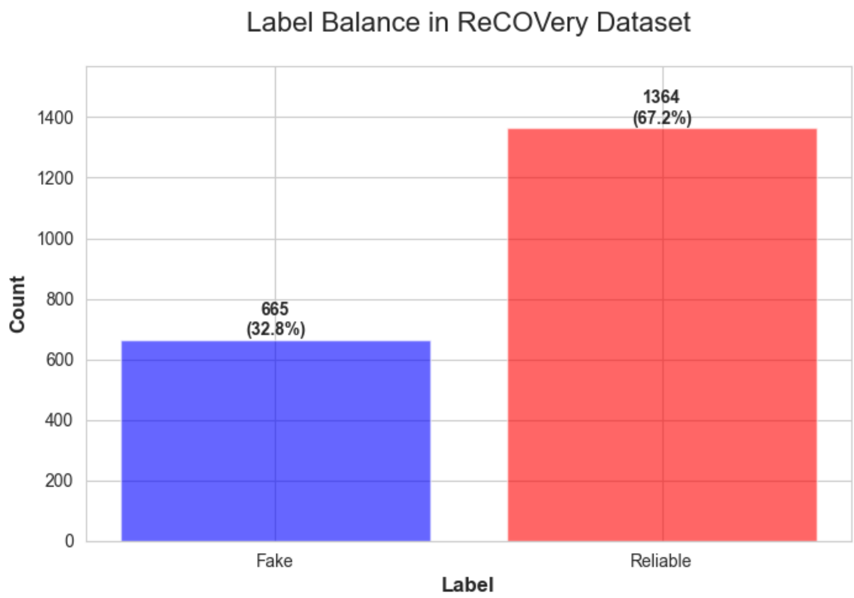


Figure 3.1 Distribution of labels in the ReCOVery dataset

Since this thesis utilizes transformer-based architectures, token length becomes an important factor during training and evaluation. Transformer models operate with a fixed input length, meaning that articles exceeding the selected token limit must be truncated. Consequently, parts of longer articles are discarded and are not considered during feature extraction or prediction.

The token length distribution for the ReCOVery dataset using the BERT tokenizer is shown in Figure 3.2. The figure presents a histogram of article lengths measured in tokens together with several selected input length limits. Larger token windows allow the model to process more information from each article, but they also significantly increase computational requirements and training time.

In this work, the maximum sequence length was limited to 512 tokens. Although larger context windows such as 1024 tokens were considered, their computational cost was too high for the available computational resources. A similar simplification strategy was also applied in [1] in order to keep the experiments computationally feasible.

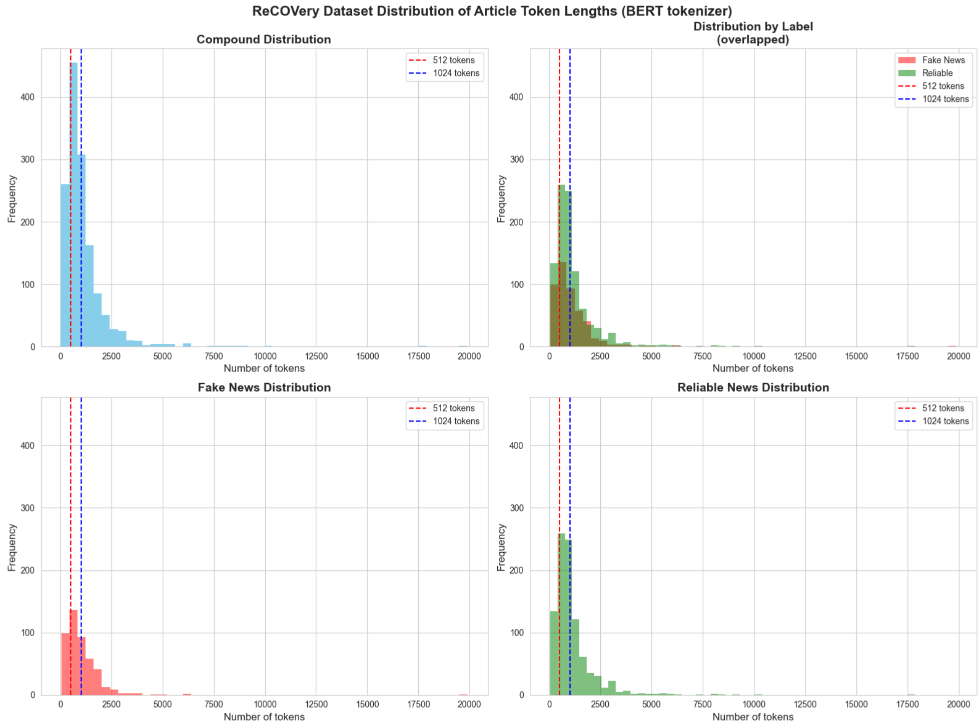


Figure 3.2 Distribution of token lengths in the ReCOVery dataset using the BERT tokenizer

The presented label distribution and token length analysis provide important information about the characteristics of the dataset. Compared to the other datasets used in this thesis, the ReCOVery dataset contains relatively long articles, in some cases exceeding 5000 tokens. A sequence length of 512 tokens therefore covers less than half of articles.

As a result, the transformer models used in this work process only partial information from longer texts, typically focusing on the beginning of the article. This represents a significant limitation of the current experimental setup and may negatively affect the models' ability to detect disinformation patterns appearing later in the text.

3.2 ISOT Dataset

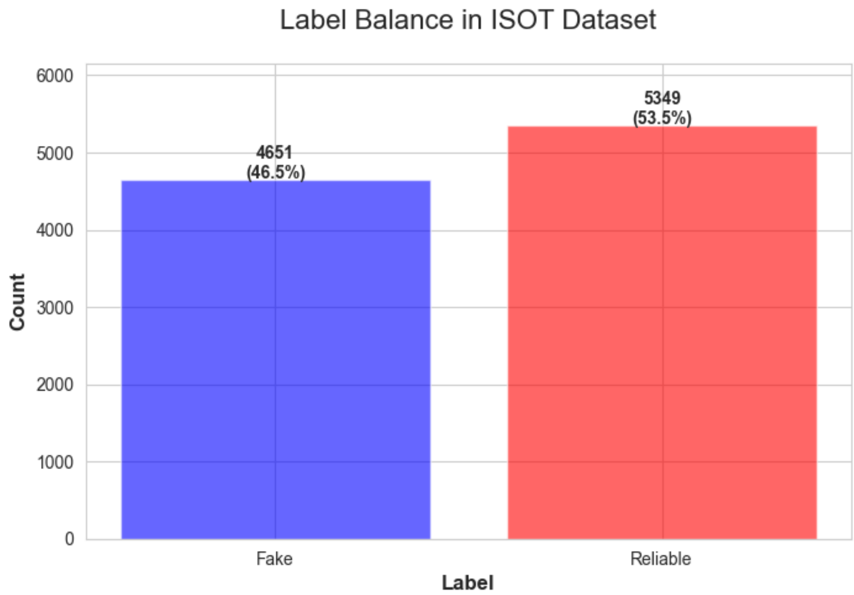
The ISOT Fake News Dataset is a widely used benchmark dataset for fake news classification tasks. It contains more than 44,000 articles, where reliable news articles were obtained from the Reuters news agency, while fake news articles were collected from unreliable sources and verified using fact-checking resources such as Politifact and Wikipedia . The dataset primarily focuses on political news published during 2016 and 2017, with a significant emphasis on topics related to former United States President Donald Trump and his policies [1].

In the previous work by Jan Flajžík [1], several preprocessing and simplification steps were applied to the original dataset:

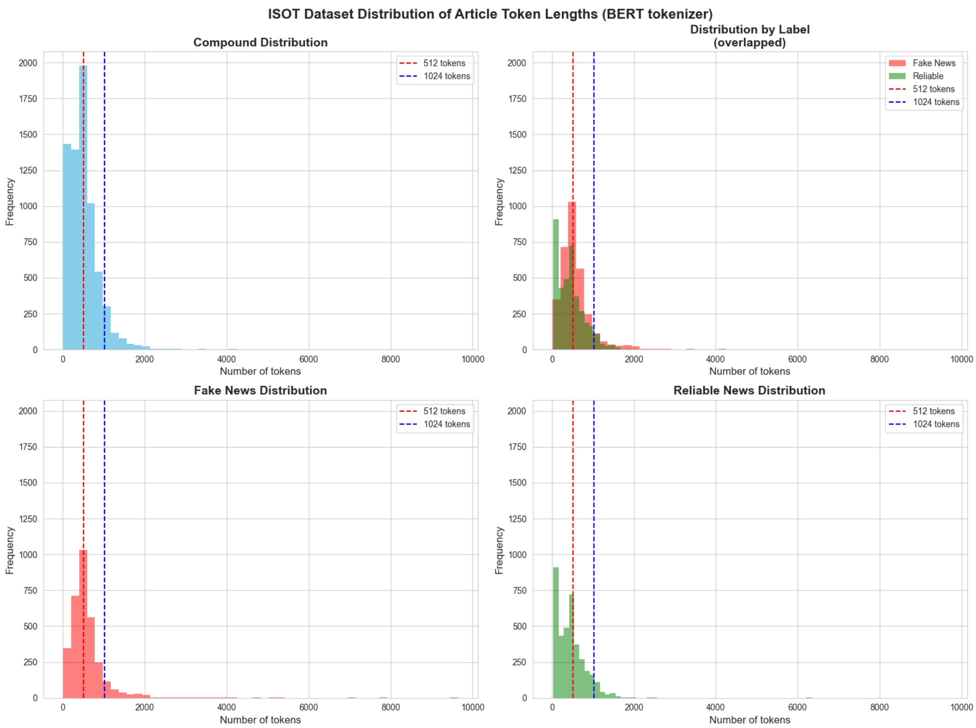
- the dataset was reduced to 10,000 articles in order to decrease the size difference between ISOT and the remaining datasets,
- articles published in 2016 were removed,
- the `subject` attribute was excluded because the categories of fake and reliable news did not overlap, making the classification task artificially simple,
- the resulting dataset remained relatively balanced, containing approximately 53% reliable news and 47% fake news.

The label distribution of the resulting dataset is shown in Figure 3.3, while the token length distribution using the BERT tokenizer is shown in Figure 3.4.

Compared to the ReCOVery dataset, the ISOT dataset contains substantially more articles, while the average article length is generally shorter. As a result, the selected maximum sequence length of 512 tokens covers a considerably larger portion of the dataset, allowing transformer models to process a greater amount of information from each article.



■ **Figure 3.3** Distribution of labels in the ISOT dataset



■ **Figure 3.4** Distribution of token lengths in the ISOT dataset using the BERT tokenizer

3.3 EUvsDisinfo Dataset

The EUvsDisinfo dataset used in the previous work by Jan Flajžík [1] was constructed using data from the EUvsDisinfo project operated by the European External Action Service's East StratCom Task Force. The initiative was established in 2015 with the goal of monitoring and documenting disinformation campaigns associated with the Russian Federation.

Several modifications were applied in order to create a suitable dataset for binary fake news classification experiments [1]:

- instead of full articles, short English summaries describing disinformation cases were used,
- reliable news summaries were collected from *The Guardian* to create a balanced counterpart dataset,
- the resulting dataset contains 2,139 text samples with an almost balanced distribution between reliable and unreliable news,
- because the dataset consists of summaries rather than full articles, the length of each sample is relatively short.

According to [1], shorter textual inputs appeared to be more suitable for certain neural architectures such as LSTM models, which achieved better results on this dataset than on datasets containing longer articles. The use of summaries may also better reflect modern online media consumption, where users often interact primarily with short-form textual content.

The label distribution of the dataset is shown in Figure 3.5, while the token length distribution obtained using the BERT tokenizer is shown in Figure 3.6.

Compared to the previously discussed datasets, the EUvsDisinfo dataset contains significantly shorter text samples. As a result, the selected transformer input window of 512 tokens is sufficient to process the complete text for all samples without truncation.

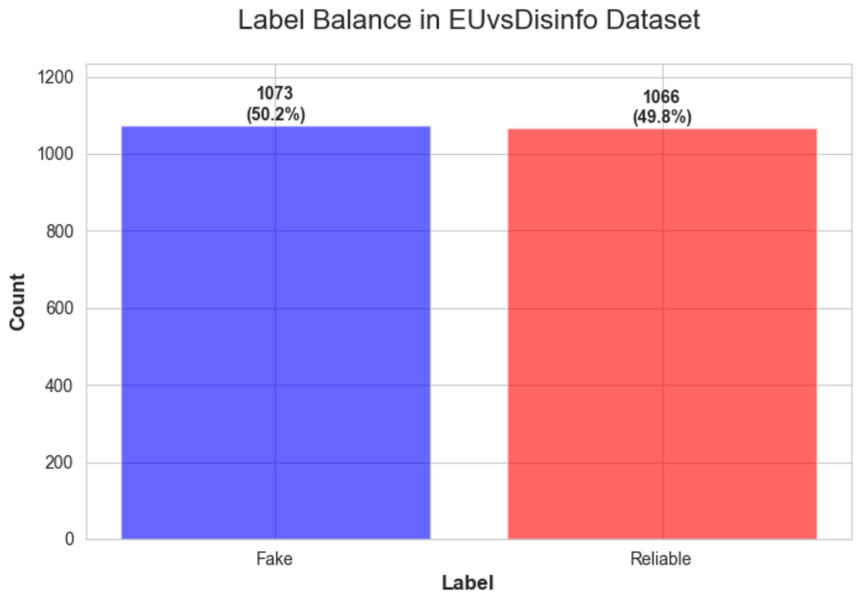


Figure 3.5 Distribution of labels in the EUvsDisinfo dataset

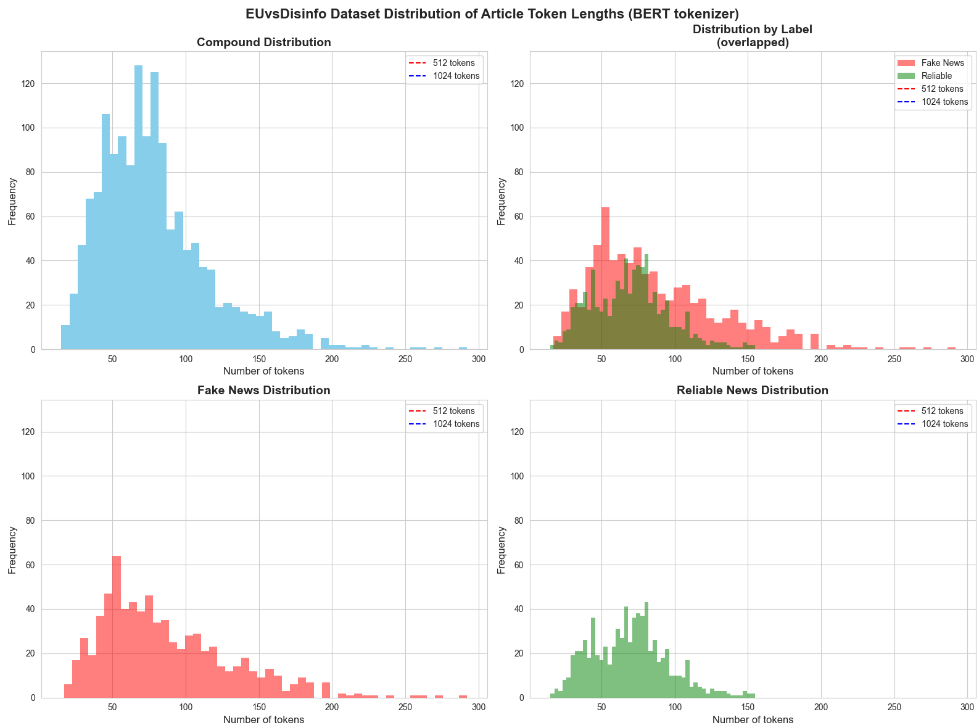


Figure 3.6 Distribution of token lengths in the EUvsDisinfo dataset using the BERT tokenizer

The token length analysis shows that the text samples in this dataset are relatively short. Since the vast majority of samples remain below the selected limit of 512 tokens, the transformer models are able to process the complete summaries without truncation. Consequently, the models can utilize the full textual information available in the dataset during both training and evaluation.

3.4 Czech Dataset

The Czech dataset used in [1] was created specifically for fake news detection experiments focused on the Czech media environment. The dataset primarily targets disinformation distributed through chain emails, which represent a common channel for spreading misleading information in the Czech Republic.

The dataset consists of both fake and reliable textual sources:

- 405 fake news texts were obtained from the Eldariel database maintained by the citizen initiative *Čeští elfové*,
- approximately 30% of the fake content consists directly of chain email texts, while the remaining samples contain articles referenced within those emails,
- reliable news articles were collected from the Czech public television news portal *ČT24*,
- the resulting dataset contains 995 text samples with an approximate ratio of 59% reliable news and 41% fake news.

The collected texts focus primarily on international political topics related to Russia and Ukraine during 2022 and the beginning of 2023. During dataset preparation, entries containing only images or links without sufficient textual content were removed.

Because Czech natural language processing requires language-specific preprocessing tools, the original implementation used several Czech-oriented resources and libraries, including Czech stemming tools, multilingual lemmatization libraries, and pretrained Czech language embeddings.

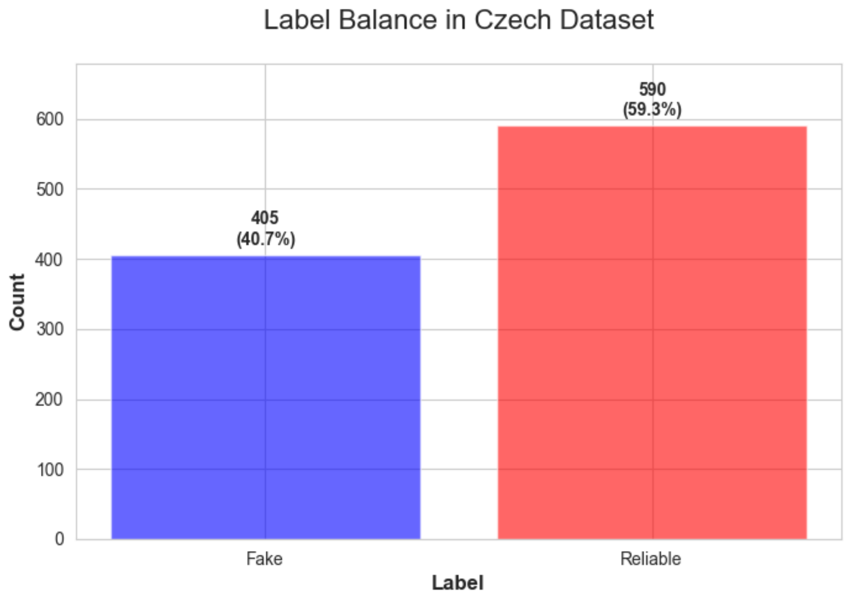


Figure 3.7 Distribution of labels in the Czech dataset

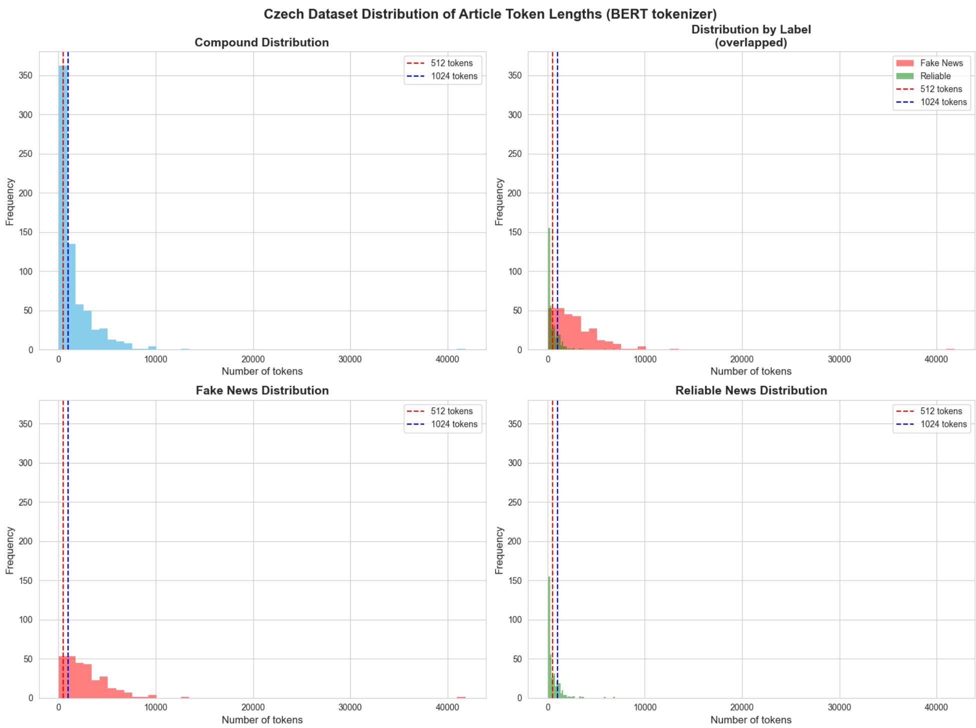


Figure 3.8 Distribution of token lengths in the Czech dataset using the BERT tokenizer

3.5 Summary

The dataset analysis highlights one of the main limitations of the experimental setup used in this thesis. Although transformer-based architectures are capable of processing complex textual information, the fixed input length restriction prevents the models from utilizing the full content of many long articles.

A more suitable approach could involve splitting longer articles into multiple segments and adapting the training process accordingly so that the entire text can be processed. Such an approach would likely provide richer contextual information to the models, although it would also increase computational requirements and implementation complexity.

The experiments conducted in this thesis therefore primarily evaluate how transformer-based architectures perform under constrained conditions, where only partial textual information is available for many samples. This limitation may lead the models to rely on more superficial textual patterns, potentially reducing generalization performance and increasing the risk of overfitting.

Chapter 4

Implementation and Experiments

This chapter describes the implementation of the experimental framework used in this thesis, including the selected tools and libraries, the integration of experiment tracking, and the methodology used for model training and evaluation.

4.1 Used Tools and Libraries

The implementation developed in this thesis relies on several widely used machine learning and natural language processing libraries:

- MLflow – an experiment tracking framework used for monitoring training runs, logging metrics, storing artifacts, and comparing model configurations through a graphical user interface.
- NLTK – a natural language processing library used mainly for selected preprocessing operations and alternative tokenization approaches.
- Transformers – the Hugging Face Transformers framework used for downloading, configuring, and training pretrained transformer-based models. All transformer architectures used in this thesis were implemented using this library.
- PyTorch – the deep learning framework used as the computational backend for model training and inference.
- Pandas and NumPy – libraries used for dataset manipulation, preprocessing, and general numerical operations.

4.2 Integration with MLflow

MLflow is an open-source framework designed for experiment tracking and machine learning workflow management. Training transformer-based models typically requires a large number of experiments involving different preprocessing pipelines, hyperparameter configurations, and model architectures. In such scenarios, systematic tracking of experiments becomes important for reproducibility, comparison of results, and debugging.

In this thesis, MLflow was integrated into the training pipeline so that all experiments, metrics, parameters, and artifacts were automatically logged for later analysis.

The greatest advantage of MLflow is its support for debugging and experiment inspection. Training deep learning models is often computationally expensive and susceptible to implementation or preprocessing errors. Logging intermediate metrics and artifacts makes it easier to identify inconsistencies and detect potential problems during training.

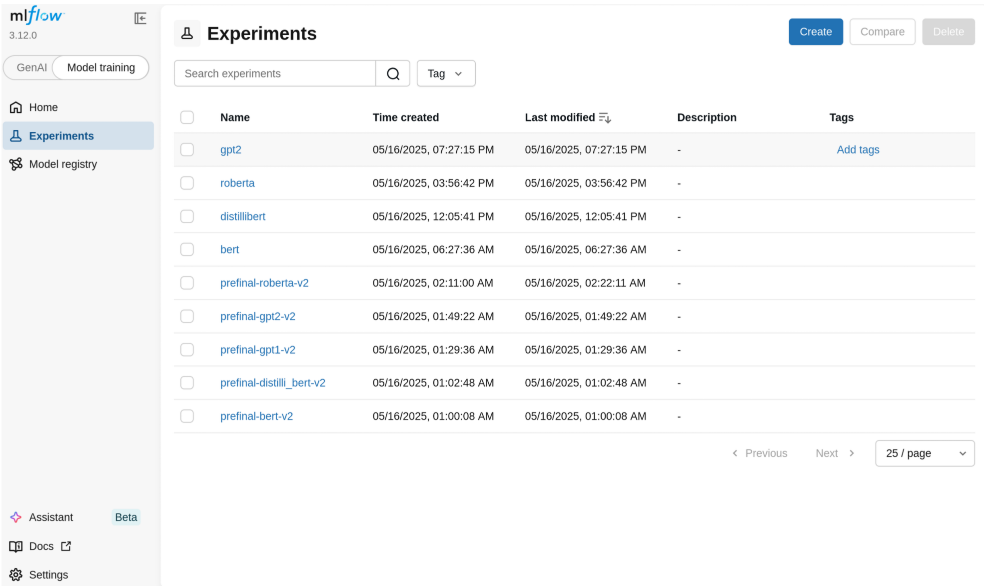
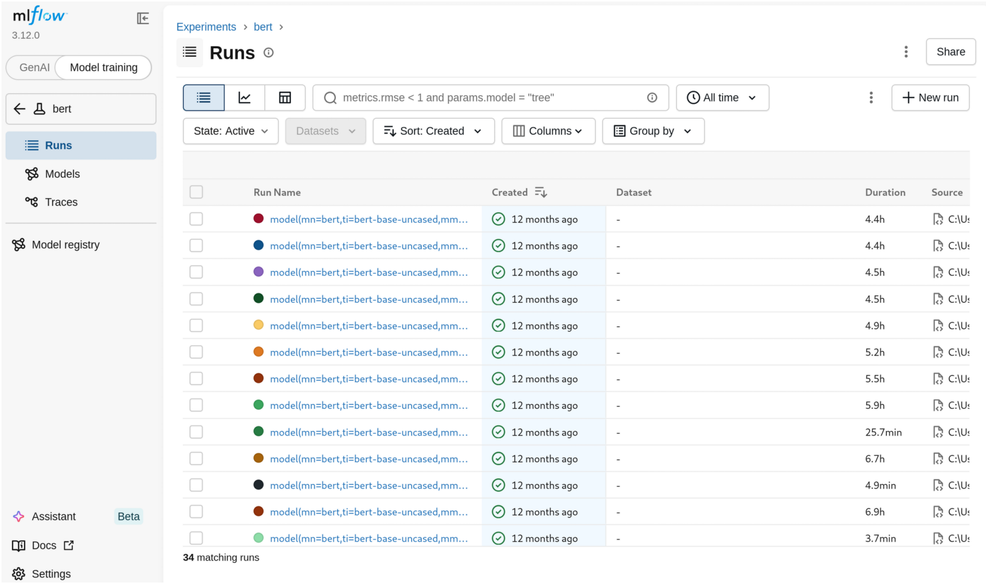


Figure 4.1 MLflow interface displaying recorded experiments

Figure 4.1 shows the MLflow experiment overview interface used during development. In this work, each transformer architecture was organized as a separate experiment, while individual runs within the experiment represented different hyperparameter configurations and preprocessing combinations.

MLflow provides a flexible interface for tracking training metadata and allows customization of displayed parameters and metrics. Besides standard training statistics, additional information such as training duration and preprocessing configurations can also be recorded.

The framework also enables continuous monitoring of evaluation metrics



■ **Figure 4.2** Recorded runs for the BERT experiment in MLflow

during training, making it possible to analyze model behavior across epochs and identify unstable or problematic runs early in the experimentation process. Custom metrics can also be added directly from the training code when needed.

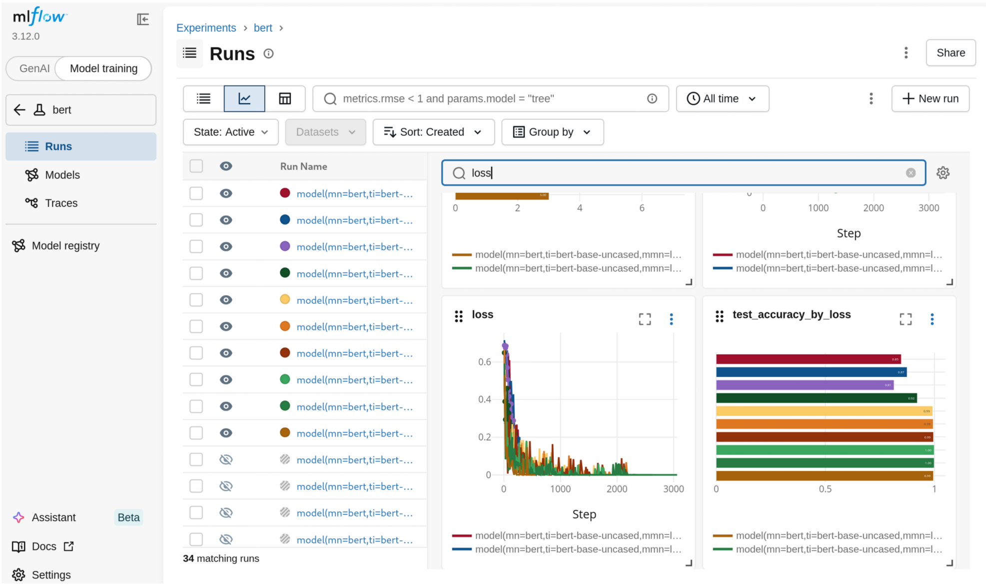


Figure 4.3 Example of tracked training metrics in MLflow

Each experiment run additionally stores the corresponding training parameters and random seeds used during execution. Many parameters are logged automatically by the Transformers framework, while additional custom information can also be recorded manually. This setup improves experiment reproducibility. Although full reproducibility is difficult to achieve in deep learning environments due to nondeterministic GPU operations and framework-level randomness, this work attempts to maximize reproducibility by consistently storing random seeds and experiment configurations.

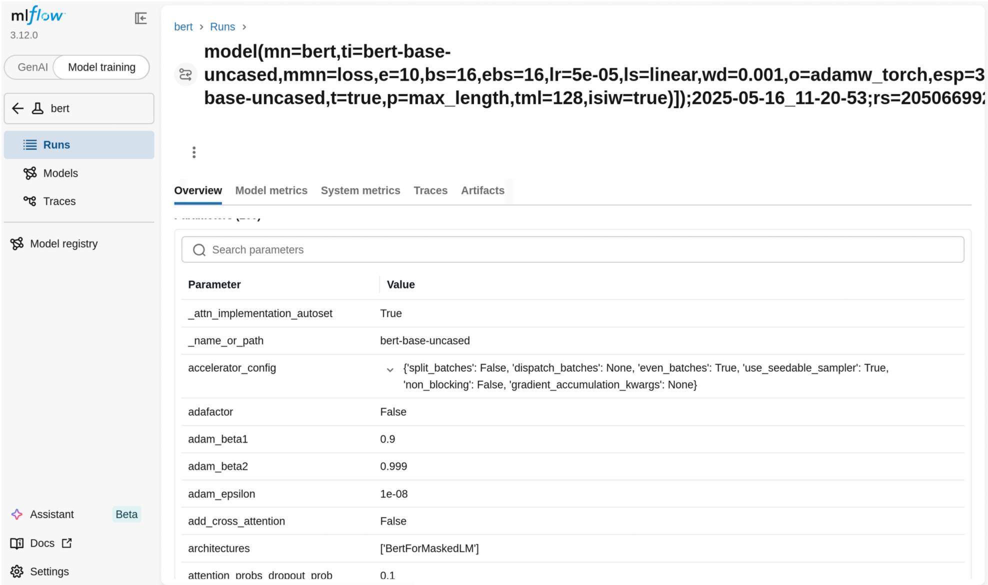


Figure 4.4 Detailed information recorded for a single MLflow run

MLflow also supports artifacts storage, including trained model checkpoints and intermediate preprocessing outputs. During experimentation, this functionality proved useful for debugging preprocessing pipelines and verifying correct data ingestion into the models. The ability to inspect intermediate artifacts and recorded metrics significantly simplified the debugging process and improved the overall reliability of the experimental workflow.

4.3 Experiments

This section describes the experimental workflow used in this thesis, including the training procedure, preprocessing pipeline configuration, and evaluation methodology.

4.3.1 Training

The training pipeline was implemented using the Hugging Face Transformers framework together with PyTorch. The implementation relies primarily on the standard model and trainer abstractions provided by the framework, while extending them with additional functionality related to experiment logging and artifact tracking through MLflow.

The developed implementation is designed in a modular manner using abstract interfaces, allowing different preprocessing pipelines, models, and training configurations to be integrated into a unified experimental workflow.

Before training, a preprocessing pipeline is configured for each experimental run. The pipeline determines which preprocessing techniques are applied to the dataset before tokenization and model training.

```
# Example of preprocessing pipeline configuration for a training run

pipeline = PreprocessingPipeline(
    name='noise_reduction_with_stemming',
    iterable=[
        NoiseReduction(),
        NLTKTokenizer(),
        Stemming(),
        Encoder(
            is_split_into_words=True,
            truncation_max_length=max_tokens
        )
    ]
)
```

Each preprocessing step automatically stores intermediate artifacts in MLflow, making it possible to inspect preprocessing outputs and verify the correctness of the pipeline during experimentation.

After preprocessing and tokenization, the encoded datasets are passed to the transformer training framework. Since the overall training procedure follows the standard workflow provided by the Transformers library, the complete implementation details are omitted.

As discussed in the dataset analysis chapter, transformer models in this work process a maximum of 512 tokens per sample. Any remaining tokens exceeding this limit are truncated and are therefore not included during training or inference.

Regarding dataset splitting, 70% of the samples were used for training, 15% for validation, and the remaining 15% for testing. The validation set was used to monitor model performance during training and to select the best-performing checkpoint based on validation metrics. The selected checkpoint was then evaluated on the test set to obtain the final reported results.

4.3.2 Evaluation

Evaluation of transformer-based architectures differs from the evaluation of simpler machine learning models because the models are trained over multiple epochs and their performance changes throughout the training process.

For each training run, evaluation metrics were recorded after every epoch. The final model checkpoint was selected according to the best validation accuracy achieved during training. In practice, this means that the checkpoint corresponding to the epoch with the highest validation accuracy was chosen as the final model for evaluation.

Several evaluation metrics were monitored during experimentation, including accuracy, precision, recall, F1-score, ROC-AUC, false positive rate, and false negative rate. Among these metrics, accuracy was selected as the primary comparison criterion in this thesis. This choice was made mainly to maintain consistency with the previous work by Jan Flajžík [1] and to simplify comparison between the classical and transformer-based approaches.

The selected checkpoint was subsequently evaluated on the test dataset to obtain the final reported results.

4.4 Conclusion

One of the main contributions of this work is the implementation of a reusable and extensible experimental framework for transformer-based fake news classification. Training large neural architectures is computationally demanding and often requires extensive experimentation, making experiment tracking and reproducibility important aspects of the implementation.

Compared to the previous work by Jan Flajžík [1], where most experiments were implemented directly in Jupyter notebooks, this thesis introduces a more modular software structure based on reusable Python modules. Jupyter notebooks were used primarily for hyperparameter exploration and interactive monitoring of ongoing experiments.

The integration of MLflow significantly simplified experiment management, debugging, and result analysis. The ability to store intermediate artifacts, preprocessing outputs, model checkpoints, and detailed training metrics proved particularly useful during debugging and validation of the experimental pipeline.

Chapter 5

Results

This chapter presents the results obtained during the experiments and compares them with the results reported in the previous work by Jan Flajžík [1]. The results are presented separately for each dataset.

5.1 Results for the ReCOVery Dataset

The results for the ReCOVery dataset are shown in Table 5.1. Among the evaluated models, BERT achieved the highest accuracy.

In the previous work by Jan Flajžík [1], the best-performing model for this dataset was Naive Bayes with an accuracy of 83.5%. In that implementation, articles were truncated to 250 words before preprocessing. Since transformer tokenizers typically split words into multiple subword tokens, the effective amount of processed text in this thesis is not directly comparable to word-based truncation. Nevertheless, the transformer-based BERT model achieved higher accuracy despite operating on only the first 512 tokens of each article.

The improvement over the previous approach is relatively moderate. However, considering that many articles in the ReCOVery dataset significantly exceed the selected token limit, it is possible that processing longer text segments or complete articles could further improve transformer performance.

Table 5.1 Best results for the ReCOVery dataset ranked by accuracy

Model	**Accuracy**	**Precision**	Recall	**F1-Score**	**ROC-AUC**	**FPR**	FNR
BERT	0.905	0.898	0.971	0.933	0.940	0.237	0.029
DistilliBERT	0.866	0.908	0.895	0.901	0.919	0.198	0.105
RoBERTa	0.807	0.818	0.917	0.865	0.850	0.424	0.083
GPT2	0.843	0.886	0.877	0.882	0.912	0.228	0.123

The preprocessing pipeline used for the best-performing configuration consisted only of tokenization using the BERT tokenizer without additional preprocessing operations such as stemming, lemmatization, or noise reduction.

The selected hyperparameters were:

- batch size: 16,
- learning rate: 5×10^{-5},
- learning rate scheduler: linear,
- weight decay: 0,
- optimizer: AdamW.

5.2 Results for the ISOT Dataset

The results for the ISOT dataset are shown in Table 5.2. Similarly to the ReCOVery dataset, BERT achieved the highest accuracy among all evaluated models.

The previous work by Jan Flajžík [1] reported approximately 97% accuracy using a CNN-based architecture with articles truncated to 250 words. In this thesis, BERT achieved near-perfect classification performance. One possible explanation is that the ISOT articles are generally shorter than those in the ReCOVery dataset, allowing the selected 512-token window to preserve a larger proportion of the original text.

Table 5.2 Best results for the ISOT dataset ranked by accuracy

Model	**Accuracy**	**Precision**	Recall	**F1-Score**	**ROC-AUC**	**FPR**	FNR
BERT	0.995	0.995	0.995	0.995	1.000	0.006	0.005
DistilliBERT	0.989	0.987	0.991	0.989	0.997	0.014	0.009
RoBERTa	0.989	0.986	0.994	0.990	0.999	0.015	0.006
GPT2	0.990	0.989	0.991	0.990	0.999	0.011	0.009

The best-performing configuration used the following hyperparameters:

- batch size: 16,
- learning rate: 5×10^{-5},
- learning rate scheduler: linear,
- weight decay: 0.001,
- optimizer: AdamW.

The preprocessing pipeline again consisted only of BERT tokenization without additional preprocessing operations.

5.3 Results for the EUvsDisinfo Dataset

Table 5.3 presents the results for the EUvsDisinfo dataset. BERT again achieved the best overall performance.

This dataset differs substantially from the previous datasets because it consists primarily of short summaries rather than full-length articles. As discussed in the dataset analysis chapter, nearly all samples fit entirely within the 512-token limit. Consequently, the transformer models are able to process the complete textual information without truncation.

The best-performing BERT configuration achieved 100% accuracy, improving upon the 95.1% accuracy with LSTM reported in [1].

Table 5.3 Best results for the EUvsDisinfo dataset ranked by accuracy

Model	**Accuracy**	**Precision**	Recall	**F1-Score**	**ROC-AUC**	**FPR**	FNR
BERT	1.000	1.000	1.000	1.000	1.000	0.000	0.000
DistilliBERT	0.978	0.956	1.000	0.977	1.000	0.041	0.000
RoBERTa	0.991	1.000	0.983	0.991	1.000	0.000	0.017
GPT2	0.981	0.982	0.982	0.982	0.998	0.019	0.018

The selected hyperparameters for the best-performing model were:

- batch size: 16,
- learning rate: 5×10^{-5},
- learning rate scheduler: linear,
- weight decay: 0.001,
- optimizer: AdamW.

The preprocessing pipeline again consisted only of tokenization using the BERT tokenizer.

5.4 Results for the Czech Dataset

The results for the Czech dataset are shown in Table 5.4. Unlike the English-language datasets, the Czech dataset produced less consistent results across the evaluated transformer architectures.

Interestingly, DistilBERT achieved better performance than the full BERT model despite being substantially smaller. Across multiple experimental runs 5.1, DistilBERT consistently produced stronger results than BERT under various hyperparameter configurations.

The experiments also suggest that preprocessing techniques had only a limited influence on the final performance. Both minimal preprocessing pipelines and more extensive preprocessing configurations produced both strong and weak results across different runs.

Another notable observation is that the Naive Bayes model reported in [1] achieved an accuracy of 89.9%, outperforming the BERT model evaluated in this thesis. At the current stage of experimentation, no clear explanation for this behavior has been identified.

Table 5.4 Best results for the Czech dataset ranked by accuracy

Model	**Accuracy**	**Precision**	Recall	**F1-Score**	**ROC-AUC**	**FPR**	FNR
BERT	0.847	0.811	0.966	0.882	0.918	0.328	0.034
DistilliBERT	0.920	0.901	0.965	0.932	0.978	0.138	0.035
RoBERTa	0.613	0.613	1.000	0.760	0.775	1.000	0.000
GPT2	0.793	0.787	0.871	0.827	0.865	0.308	0.129

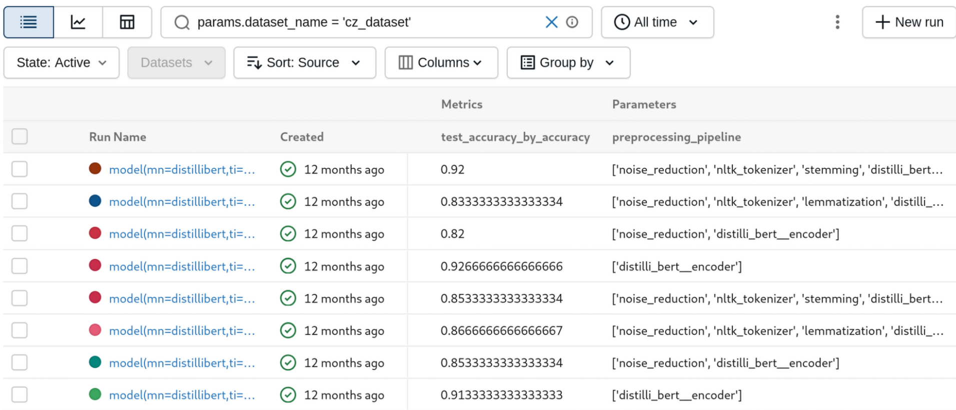


Figure 5.1 Example of DistilBERT experiment runs for the Czech dataset recorded in MLflow

Chapter 6
Conclusion

This thesis extended the previous work by Jan Flajžík [1] through the application of transformer-based architectures for fake news classification. The conducted experiments demonstrated that modern transformer models are capable of achieving strong performance across multiple datasets and, in several cases, improving upon the results obtained using classical machine learning approaches.

Another important contribution of this work is the implementation of a reusable and extensible experimental framework for transformer-based natural language processing tasks. The developed pipeline integrates preprocessing, model training, experiment tracking, and artifact management through MLflow, simplifying experimentation, debugging, and reproducibility.

The experiments also highlighted several limitations of the current approach. In particular, the fixed transformer input length represents a significant challenge for datasets containing long articles, where substantial portions of the text are truncated before processing. This limitation may prevent the models from utilizing important contextual information and can negatively affect generalization performance.

Future work could therefore focus on methods for processing longer documents more effectively, for example by splitting articles into multiple segments or by utilizing architectures designed for long-context processing. Additional experiments on larger and more diverse datasets could also provide further insight into the robustness of transformer-based fake news classification systems.

More broadly, the results suggest that careful experimentation, modular implementation, and efficient utilization of computational resources may be more beneficial than simply increasing model or dataset size without methodological improvements.

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Contents of attached medium

The attached medium contains the source code for the implemented experiments, as well as visualization scripts, analytical notebooks, and execution notebooks for running and training the experimental models.

```
materials
├── BP Flajžík Jan .......... original thesis [1] text and sources
└── etc... ........................................ other materials
workspace
├── mlruns ................................ logs used in the MLflow
├── notebooks ........................ training and visualizations
├── sources .............................. reusable python modules
│   ├── experiments ............ package with experiment runs logic
│   ├── local_datasets ........ datasets analysis and preprocessing
│   ├── models ....................... machine learning models logic
│   ├── __init__.py
│   └── utils.py
└── tests
README.md
.gitignore
requirements.txt ........................... packages to install
```